SS-R-1781
October 2, 1972
SUPERSEDING
MIL-R-3472A
February 10, 1972

FEDERAL SPECIFICATION

# ROOF-COATING, ASPHALT-BASE EMULSION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers asphalt-base emulsions for roof applications.

1.2 Classification. The emulsion shall be of the following types (see 6.2).

Type I - Asbestos filler, mineral colloid emulsifying agents.
Type II - Mineral fillers, chemical emulsifying agents.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Federal Specifications:

TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.
PPP-P-704 - Pails, Metal (Shipping Steel, 1 through 12 Gallon).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads (40 inches by 48 inches, 4 way Partial and 4 way Pallets).

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting Officer.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:14Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D140 - Sampling Bituminous Materials.
D1167 - Testing Asphalt-Base Emulsions for use as Protective Coating for Built-Up Roofs.
D1227 - Asphalt-Base Emulsions for use as Protective Coatings for Built-Up Roofs.
D2939 - Bituminous-Base Emulsions for use as Protective Coatings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for references from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The emulsion shall be a homogenous mixture either of the mineral colloid or chemical type, compounded from petroleum asphalt and mineral fillers or fibers. Dispersing agents must be nonflammable. Petroleum or coal tar derivatives shall not be used as solvents.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of a good commercial quality entirely suitable for the purpose. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.3 Types.

3.3.1 Type I. Type I asphalt-base emulsion shall conform to the composition and performance requirements of type I asphalt-base emulsion of ASTM D1227, except that the maximum weight shall be 9.0 pounds.

3.3.2 Type II. Type II asphalt-base emulsion shall conform to the composition and performance requirements of type II asphalt-base emulsion of ASTM D1227.

3.4 Uniformity. The emulsion shall be homogeneous and shall show no separation of water or coagulation of the asphalt base. The emulsion shall evidence no settling or packing in the container that cannot be overcome by moderate stirring.

3.5 Consistency. The consistency shall be suitable for application by spray, brush, or mop, at temperatures above freezing, without heating or thinning, at a rate of approximately 2-1/2 gallons per 100 square feet. The emulsion shall bond to all clean surfaces, whether damp or dry.

3.6 Marking. Each container of emulsion shall have the manufacturer's name or trademark stamped, or otherwise permanently marked upon the container.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:14Z
Check the source to verify that this is the current version before use.

3.7 Special marking. Each container of emulsion shall be stemciled or labeled in a clear, legible manner with the following words:

Asphalt-Base Roof-Coating - Do Not Allow to Freeze.
Do Not Store at a Temperature Below 40° F.
PERMANENTLY DAMAGED IF FROZEN.

3.8 Workmanship. The quality of workmanship shall be such as to produce an emulsion that is in accordance with the requirements of the specification and be free from harmful defects which would interfere with use of material in the application for which it is intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Lot. A lot shall consist of each type of emulsion offered for delivery at one time.

4.3.2 Sampling for tests. Samples for tests shall be in accordance with level S-1 of MIL-STD-105 or ASTM D140.

4.4 Test methods. Each sample selected in accordance with 4.3.2 shall be tested by the appropriate section of ASTM D1167 or D2939. Failure of the sample to pass any test shall be cause for rejection.

4.5 Examination of preparation for delivery. An examination shall be made to determine that markings, materials, workmanship and contents comply with the requirements of section 5. The sample unit shall be one shipping container fully packed, selected just prior to the closing operations. The closed shipping containers shall be examined for closure defects. The inspection level shall be S-2 and the AQL shall be 4.0 expressed as defects per hundred units.

TABLE I. Classification fo preparation for delivery defects.

| Examine | Defects |
|---|---|
| Marking (interior and exterior container) | Incorrect; incomplete; illegible; omitted; of improper size; location, sequence, or method of application. |
| Materials | Any nonconforming component, component missing, damaged or otherwise defective. |
| Workmanship | Inadequate application of components such as incomplete closure of container flaps loose or inadequate sealing strapping or stapling when required. |
| Contents | Number of unit containers in the shipping container more or less than required. |
| Containers | Not as specified, not close-fitting when required for level A or B. |

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:14Z
Check the source to verify that this is the current version before use.

5.1.1 Level A. Unless otherwise specified (see 6.2), the roof coating, furnished in 5 gallon quantities, shall be packed in a pail conforming to PPP-P-704, type II, class 1, 2, or 3.

5.1.2 Level B. The roof-coating, furnished in 5 gallon quantities, shall be packed in a 5-gallon, 29 gage metal pail with lug cover and wire bail. Each cover shall have a flowed in or other suitable gasket uniformly distributed about the circumference of the inner edges of the cover. The gasket material shall be of sufficient thickness to provide an adequate seal. The wire bail shall be fabricated from No. 9 W & M gage with a nominal thickness of 0.148 and shall be zinc coated or otherwise coated to resist corrosion. All exterior surfaces of the pail shall have a commercial coating.

5.1.3 Level C. The roof-coating shall be packed so as to insure arrival at destination in a satisfactory condition in containers that comply with the Uniform Freight Classification rules or the National Motor Freight Classification rules, as applicable.

5.2 Palletization. When specified (see 6.2), containers shall be palletized in accordance with MIL-STD-147.

5.3 Marking. In addition to any marking required by the procurement document, the roofing coating shall be marked in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as applicable.

6. NOTES

6.1 Intended use. The asphalt-base emulsion is intended as a protective coating for smooth-surface asphalt roofs having inclines of not less than 1/8 inch to the foot.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type of emulsion required (see 1.2).
(c) Level of packaging and level of packing required (see 5.1).
(d) When palletization is required (see 5.2).

Preparing activity:

GSA-FSS



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:14Z
Check the source to verify that this is the current version before use.